



Questions	04
Pages	03

Trincomalee Campus, Eastern University, Sri Lanka
Faculty of Applied Science
Department of Computer Science
Bachelor of Computer Science (2020/2021)
Year I Semester II (July 2023)
CO1225 Computer Architecture

Answer All Questions

Time: Two hours

Q1)

a) Convert the following hexadecimal numbers to binary and octal numbers:

i) $764A_{16}$;

ii) $8945FA6B_{16}$;

iii) $18AF67B_{16}$.

[12 Marks]

b) Convert the following binary numbers to decimal equivalents.

i) 11010101_2 ;

ii) 11010.110_2 ;

iii) 010011110.1010_2 .

[06 Marks]

c) Convert the decimal number 67.567 to binary number (should be truncated to 5 digits).

[05 Marks]

d) Compute the following decimal arithmetic operation using 8-bit 1's complement:

i) $-16+78$;

ii) $-90-8$;

iii) $13-26$.

[12 Marks]

e) Compute the following decimal arithmetic operation using 8-bit 2's complement:

i) $15-7$;

ii) $-10+6$;

iii) $-23-7$.

[15 Marks]

Q2)

- a) A system is monitored using sensors. The sensors output binary values corresponding to physical conditions, as shown in the table:

Parameter	Description of parameter	Binary value	Description of condition
P	Oil pressure	1	Pressure ≥ 3 bar
		0	Pressure < 3 bar
T	Temperature	1	Temperature $\geq 200^{\circ}\text{C}$
		0	Temperature $< 200^{\circ}\text{C}$
R	Rotation	1	Rotation ≤ 1000 rpm
		0	Rotation > 1000 rpm

The outputs of the sensors form the inputs to a logic circuit. The output from the circuit X, is 1 if any of the following three conditions occur:

If (Oil pressure ≥ 3 bar and Temperature $\geq 200^{\circ}\text{C}$) or

(Oil pressure < 3 bar and Rotation > 1000 rpm) or

(Temperature $\geq 200^{\circ}\text{C}$ and Rotation > 1000 rpm)

- i) Draw a block diagram for the circuit showing all inputs and outputs of the Circuit. [03 Marks]
- ii) Draw a truth table for the above system. [05 Marks]
- iii) Write Boolean expression for the output. [10 Marks]
- iv) Simplify the above (ii) driven equation using Boolean algebra. [10 Marks]
- v) Draw the circuit using basic logic gates. [10 Marks]
- b) Simplify the following Boolean expressions using the "Boolean Algebraic" methods.
- i) $F = (A + B)(A + \bar{B})(\bar{A} + C)$
- ii) $F = \bar{A}B + B\bar{C} + BC + A\bar{B}\bar{C}$ [12 Marks]

Q3)

- a) Draw the block diagram and condensed truth table for 4 to 1 *multiplexer*. [10 Marks]
- b) Draw the logical circuit of the 2 to 4 *Line Decoder*. [06 Marks]
- c) Draw the diagram for *Von-Neumann Computer Architecture*. [10 Marks]

d) List and explain the operations performed by "Registers".

[09 Marks]

e) Write short notes for the followings:

i) Cache Memory ;

ii) Main Memory;

iii) Secondary memory.

[15 Marks]

Q4)

a) Briefly describe *Flynn's taxonomy* for classifying computer architectures.

[08 Marks]

b) Compare and contrast the functional operation of the "Reduced Instruction Set Architecture" (RISC) versus "Complex Instruction Set Architecture" (CISC).

[08 Marks]

c) Explain the function of a *program counter*.

[06 Marks]

d) Describe the term "Shared Memory"?

[06 Marks]

e) What do you mean by "Pipeline" in computer architecture?

[06 Marks]

f) Briefly describe the following addressing modes:

i) Immediate addressing mode;

ii) Direct addressing mode;

iii) Indirect addressing mode;

iv) Indexed addressing mode.

[16 Marks]